Quantification: 682_z08
=======================
PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
----

measure	value
lacuna count	11
interior lacuna count	9
frame-edge lacuna count	2

Interior lacunae only: mean, median and sample SD (n = 9)
----

measure	kind	mean	median	sd	units
area_px2	lacuna shape	2599.88	2671.5	1127.64	px^2
major_axis_length_px	lacuna shape	103.523	92.8874	29.9285	px
minor_axis_length_px	lacuna shape	33.6052	36.6597	6.7292	px
aspect_ratio	lacuna shape	3.087	2.9214	0.6438	unitless
eccentricity	lacuna shape	0.9401	0.9395	0.0221	unitless
solidity	lacuna shape	0.8296	0.873	0.1073	unitless
roots_count	headline	13.6667	14	5.8523	count
ring_length_r30_px	headline	492.778	500	213.97	px
ring_length_r60_px	reported	1074.78	1039	482.002	px
owned_length_px	ownership-dependent	2994.98	2202.2	2235.55	px
edge_count	ownership-dependent	146.667	106	107.419	count
mean_edge_length_px	ownership-dependent	20.0508	20.3535	2.9219	px
perimeter_px	normalised	253.461	234.894	106.721	px
ring_area_r30_px2	normalised	9783.33	9374	2822.56	px^2
ring_area_r60_px2	normalised	23226.1	23858	6006.58	px^2
in_frame_fraction_r30	normalised	1	1	0	unitless
in_frame_fraction_r60	normalised	0.9809	1	0.0441	unitless
ring_density_r30	normalised	0.0487044	0.0522712	0.0135516	px^-1
ring_density_r60	normalised	0.0448407	0.0454615	0.0143901	px^-1
roots_per_100px_perimeter	normalised	5.5684	5.4625	1.2117	per 100 px
ring_attached_length_r30_px	network v2	454.333	472	217.158	px
ring_attached_length_r60_px	network v2	938	882	515.286	px
ring_length_w_r30_px	network v2	558.35	557.747	238.233	px
ring_length_w_r60_px	network v2	1216.47	1168.95	537.41	px
sholl_crossings_r10	network v2	12	10	6.0208	count
sholl_crossings_r20	network v2	14.2222	14	5.1424	count
sholl_crossings_r30	network v2	15.3333	14	6.6144	count
loop_count	loops, report only	14.2222	11	12.1735	count
loop_edge_share	loops, report only	0.3149	0.3774	0.1446	share
ring_width_mean_r30_px	width	6.9773	7.2323	0.6858	px

Per field
----

measure	value	units
analysed_area_px2	1.01768e+06	px^2
total_skeleton_length_px	37374	px
canalicular_length_density_per_px	0.0367246	px^-1
junction_count	886	count
junction_density_per_px2	0.0008706	px^-2
lacuna_count	11	count

lacunae_per_px2	1.081e-05	px^-2
skeleton_component_count	262	count
mean_component_length_px	142.649	px
median_component_length_px	17.5	px
field_length_density_w_per_px	0.0413253	px^-1
field_density_without_flagged_per_px	0.0378157	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	950	px
unattached_component_count	54	count
canalicular_length_density_attached_per_px	0.0367246	px^-1
field_length_density_with_unattached_per_px	0.037658	px^-1
loop_count	184	count
loop_edge_share	0.3711	share
edge_count	1978	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.552	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.198	px
width_pixels_used	35649	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	9321	210.577	69.7066	3.0209	0.9436	0.684	1.3112	31.8256	428.162	True
2	2563	90.1026	36.7546	2.4515	0.913	0.9144	1.5381	342.687	87.595	False
3	3496	113.895	40.712	2.7976	0.9339	0.8552	-1.2503	434.254	701.917	False
4	4123	159.751	36.5648	4.369	0.9735	0.7865	1.4743	433.966	210.1	False
5	1521	89.609	30.1271	2.9744	0.9418	0.641	-1.2077	520.501	556.951	False
6	862	61.0771	21.2942	2.8683	0.9373	0.7077	-1.4443	545.484	438.164	False
7	547	50.4784	17.0874	2.9541	0.941	0.8116	-0.4973	672.987	1013.45	True
8	3601	127.976	37.0473	3.4544	0.9572	0.8909	1.4857	713.926	152.006	False
9	1853	92.0913	27.1748	3.3889	0.9555	0.9155	-1.2655	723.731	828.852	False
10	2780	93.6835	39.1668	2.3919	0.9084	0.926	-1.1692	908.613	727.092	False
11								347.049	976.84	False

Network per lacuna

lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	16	461	1108	4602.88	229	20.0999	6.2155
2	14	506	1198	3267.87	173	18.8894	7.719
3	19	512	1150	2957.07	151	19.5832	7.2323
4	24	886	1962	4801.54	264	18.1877	7.2707
5	11	481	892	2202.2	106	20.7754	7.0941
6	9	500	1039	1933.58	95	20.3535	6.3797
7	4	212	481	653.595	24	27.2331	6.0184
8	18	685	1516	7913.4	361	21.9208	7.4282

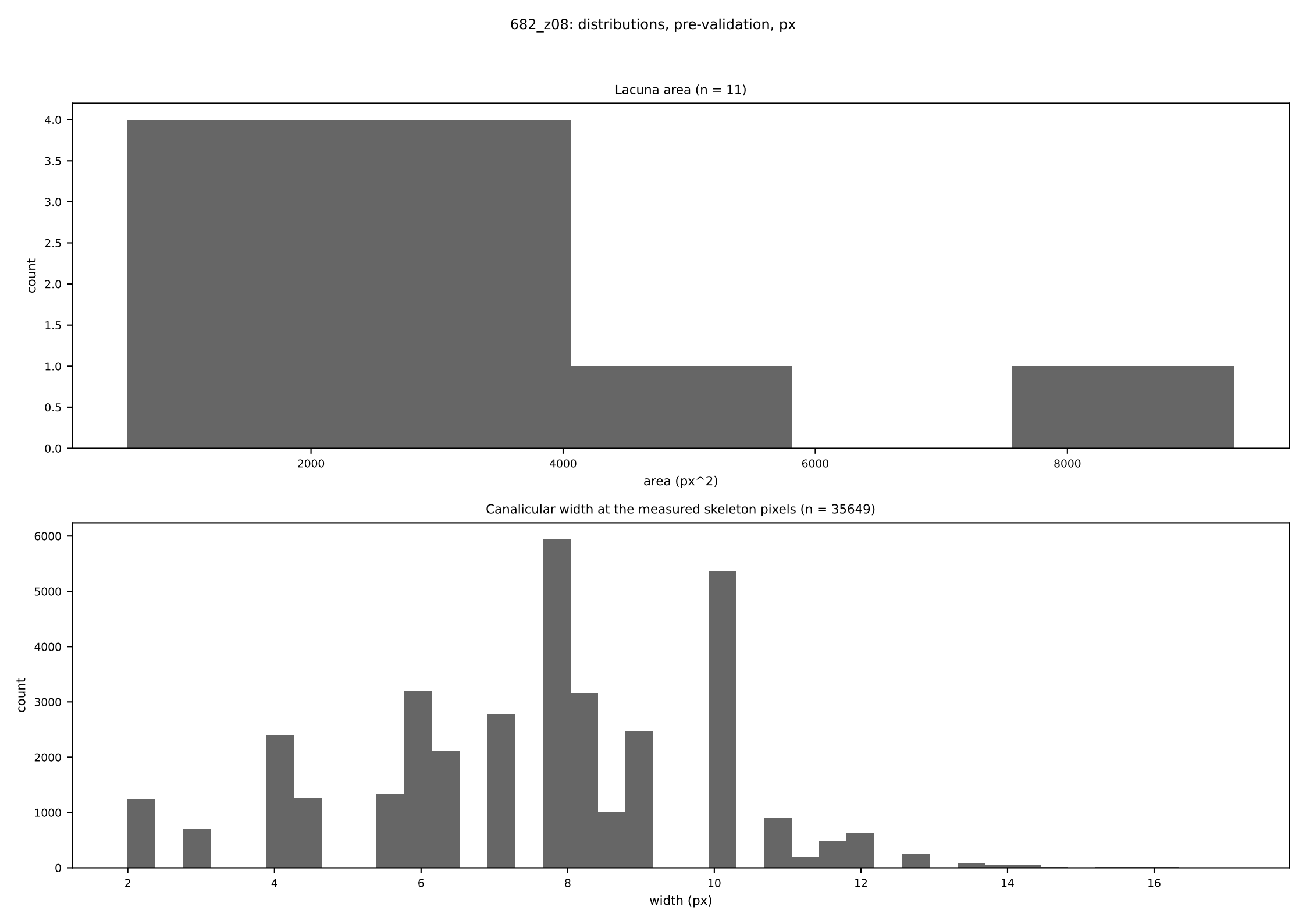
9	14	411	878	1789.29	79	22.6492	7.0137
10	8	341	806	1949.54	81	24.0684	7.236
11	6	113	232	140.297	10	14.0296	5.4219

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	769.541	14367	29344	0.6579	0.6229	0.0320874	0.037759	2.0792
2	234.894	9374	20769	1	0.9593	0.0539791	0.0576821	5.9601
3	313.948	11274	26474	1	1	0.0454142	0.0434389	6.052
4	456.961	14983	32605	1	1	0.0591337	0.0601748	5.2521
5	233.238	9202	19621	1	1	0.0522712	0.0454615	4.7162
6	166.125	7387	17424	1	1	0.0676865	0.0596304	5.4176
7	122.604	3917	10590	0.6212	0.5802	0.054123	0.0454202	3.2625
8	329.522	12020	29668	1	1	0.0569884	0.0510988	5.4625
9	222.752	9104	23858	1	1	0.045145	0.0368011	6.285
10	245.965	9693	25001	1	1	0.03518	0.0322387	3.2525
11	77.7401	5013	13615	1	0.8689	0.0225414	0.01704	7.718

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	408	759	515.546	1209.87	15	12	12
2	491	1164	570.546	1342.52	12	14	17
3	491	992	585.801	1318.93	16	14	14
4	870	1911	991.796	2210.13	23	22	27
5	443	826	547.203	1027.41	10	15	13
6	472	882	557.747	1168.95	10	14	13
7	174	415	245.865	564.299	6	7	8
8	605	1343	775.47	1681.96	17	19	22
9	356	698	471.103	1019.36	10	14	15
10	278	542	396.191	918.879	8	13	14
11	83	84	129.297	260.108	2	3	3


682_z08: distributions, pre-validation, px
Lacuna area (n = 11)
count
area (px^2)
Canalicular width at the measured skeleton pixels (n = 35649)
count
width (px)